as a helpful shorthand. For the Dirac delta δ, its action on a function $\phi \in \mathscr{T}$ is $\delta[\phi] = \phi(0)$. We write this also as

$$\langle \delta, \phi \rangle = \int \delta(z)\phi(z)\mathrm{d}\omega = \phi(0).$$

This concept of the delta function (that appears in the integral) may be motivated by introducing a sequence of continuous functions d_ε having the properties (i) $\int d_\varepsilon \mathrm{d}\omega = 1$, (ii) $\mathrm{supp}(d_\varepsilon) \to 0$, as $\varepsilon \to 0$. Then the action of the delta function can be seen as the limit of the actions of the functions d_ε. Alternatively, one may use a sequence of smooth functions $\{d_\varepsilon\}$, such as, in one dimension,

$$d_\varepsilon = \frac{1}{\sqrt{2\varepsilon\pi}} \exp\left(-\frac{z^2}{2\varepsilon}\right).$$

An example where the smooth probability density in Euclidean space is not quite the right one is in the setting of conservative (Hamiltonian) systems such as our N-body molecular system, since the evolution is restricted by invariants. The most obvious of these is the energy which we know to be a constant of motion. Therefore we need to work not on open subsets of the phase space $\mathbb{R}^m$ of our differential equations, but on lower dimensional submanifolds embedded within the phase space, e.g. the energy surface. It will be necessary to assume a density that is defined over the submanifold of constant energy. If other invariants are present, such as fixed total momentum, the discussion would need to be modified to reflect this fact.

5.3.2 *Steady States*

We are inclined to view the steady states (equilibrium solutions) of the Liouville equation as solutions of

$$\mathcal{L}_f^\dagger \rho = 0.$$

However, this requires a little care. In general, we would like to be able to apply $\mathcal{L}_f^\dagger$ to nonsmooth distributions such as Dirac delta functions for which the application of the Liouville operator does not make obvious sense (as we have not mentioned how to differentiate delta functions). Instead we will define the stationary solution in terms of integrals against test functions.

The idea of defining a distribution in terms of an integral against another function is very appropriate in the setting of molecular dynamics. We are, after all, interested in the convergence of averages of smooth functions with respect to the evolving

probability distribution. Let ϕ be a C^∞ function of the phase variables with compact support and define the time t average by

$$\bar{\phi}(t) = \frac{\int_{\mathcal{D}} \phi(\boldsymbol{q},\boldsymbol{p})\rho(\boldsymbol{q},\boldsymbol{p},t)\mathrm{d}\omega}{\int_{\mathcal{D}} \rho(\boldsymbol{q},\boldsymbol{p},t)\mathrm{d}\omega}. \tag{5.6}$$

As the flow is Hamiltonian, $\int_{\mathcal{D}} \rho(\boldsymbol{q},\boldsymbol{p},t)\mathrm{d}\omega \equiv \int_{\mathcal{D}} \rho(\boldsymbol{q},\boldsymbol{p},0)\mathrm{d}\omega$. Thus the condition for $\bar{\phi}$ to converge is just that

$$\int_{\mathcal{D}} \phi(\boldsymbol{q},\boldsymbol{p})\frac{\partial\rho}{\partial t}(\boldsymbol{q},\boldsymbol{p},t)\mathrm{d}\omega \to 0.$$

Since

$$\frac{\partial\rho}{\partial t} = \mathcal{L}_f^\dagger \rho,$$

this suggests that the condition for the invariant probability density function ρ_∞ should be

$$\int_{\mathcal{D}} \phi \mathcal{L}_f^\dagger \rho_\infty \mathrm{d}\omega = 0,$$

for all suitable functions $\phi \in C_0^\infty(\mathbb{R}^m)$. We say that ρ_∞ satisfies

$$\mathcal{L}^\dagger \rho_\infty = 0,$$

in the *weak* or *variational* sense. When such a density exists, we refer to the associated probability distribution as an *invariant distribution*. Such invariant distributions are essential tools in molecular modelling. In the case of a Hamiltonian system, we know that $\mathcal{L}_f^\dagger = -\mathcal{L}_f$, and, moreover,

$$\mathcal{L}_f H = 0,$$

so that H is a first integral. Any probability distribution whose density is a smooth function of H will also be an invariant distribution.

In particular, consider the density

$$\rho_\varepsilon(\boldsymbol{q},\boldsymbol{p}) = Z_\varepsilon^{-1} e^{-(H(\boldsymbol{q},\boldsymbol{p})-\mathrm{E})^2/2\varepsilon},$$

where ε is an arbitrary small positive constant and E is an accessible energy level. We assume that H tends to infinity sufficiently rapidly as $\|\boldsymbol{z}\| \to \infty$ so that the exponential decays sufficiently rapidly (and thus the total integral exists) and take